PROBLEM 23

Simple Pendulum (Unilateral Constraint)

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Pendulum & Liftoff

A simple pendulum consists of a mass m hanging from a string of length L and fixed at a pivot point O. The simple pendulum is equivalent to the problem of a heavy particle constrained to move along a smooth circumference. A unilateral constraint is considered, such that the particle move [sic] along the interior of the circumference and it might fall under certain conditions.

Analyze the motion of the particle when it is pushed from the lowest point of the circumference with velocity v_0. At what point does the particle leave the circumference? Is this point ever reached?

If the circumference rotates around the vertical axis with a constant angular velocity equal to ω, write down the new equations of motion. Compute the locations along the circumference in which the particle is in equilibrium.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle of mass m hangs from a string of length L fixed at the pivot O, or equivalently moves along the smooth interior of a vertical circle of radius L. In part (b) the circle rotates about the vertical diameter with the constant angular velocity ω.
- **Constraints:** the particle stays on the circle, $|\mathbf{r}| = L$ in the vertical plane. Free in space it has 3 coordinates; the plane and the circle are two holonomic constraints, so $\mathrm{CDOF} = 3 - 2 = 1$, with the angle θ measured from the downward vertical. The constraint is **unilateral**: the particle can leave the circle if the reaction would have to become negative ($N = 0$, or slack string). In part (b) the plane turns at the prescribed rate, so the constraint depends explicitly on time (rheonomic); it is solved in the inertial frame.

- **Initial motion:** pushed from the lowest point ($\theta = 0$) with speed v_0.
- **Parameters:** m [kg], L [m], g [m/s^2], v_0 [m/s], $\omega \geq 0$ [rad/s].

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with $\mathbf{k}$ vertical upward and the circle in the plane Oxz for part (a).
- **Polar basis in the plane**, with θ from the downward vertical:

$$\mathbf{e}_r = \sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{k} \ \ (\text{outward}), \qquad \mathbf{e}_\theta = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{k} \ \ (\text{along increasing } \theta)$$

- **Change of basis.** The triple $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{j}\}$ has $\mathbf{e}_r \times \mathbf{e}_\theta = -\mathbf{j}$, so it is left-handed. The right-handed polar triple is $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_3\}$ with $\mathbf{e}_3 = -\mathbf{j}$ (the third vector plays no role in the planar kinematics). Its columns in $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ give

$$[{}_0R_1] = \begin{pmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \\ -\cos\theta & \sin\theta & 0 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

Multiplying out, RR^T has diagonal entries $\cos^2 + \sin^2 = 1$ of the rotation angle (and 1) and off-diagonal entries $\pm(\cos\sin - \sin\cos) = 0$, so it is the identity.

- The angular velocity of the basis is $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{e}_3 = -\dot\theta\,\mathbf{j}$, and Poisson's formula gives $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$.
- **For part (b)**, the cylindrical basis $\{\mathbf{e}_\rho, \mathbf{e}_\varphi, \mathbf{k}\}$ is obtained from $\mathcal{B}_0$ by the simple rotation $\mathrm{R}_z(\omega t)$ about $\mathbf{k}$ (matrix $\begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\det = \cos^2\omega t + \sin^2\omega t = 1$; RR^T has diagonal entries $\cos^2\omega t + \sin^2\omega t = 1$ (and 1) and off-diagonal entries $\cos\omega t \sin\omega t - \sin\omega t\cos\omega t = 0$, so $RR^T = I$); the meridian plane of the circle is $\varphi = \omega t$ and the pair $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ is written with $\mathbf{e}_\rho$ in place of $\mathbf{i}$.

PHASE 3

Derivation

3.1 Kinematics

Part (a): planar motion

> **Pedagogical Justification:** the particle moves on a circle of constant radius, so $\mathbf{r} = L\,\mathbf{e}_r$ and, with Poisson's formula,

$$\mathbf{v} = L\dot{\theta}\,\mathbf{e}_\theta, \qquad \mathbf{a} = -L\dot{\theta}^2\,\mathbf{e}_r + L\ddot{\theta}\,\mathbf{e}_\theta$$

Part (b): rotating circle

> **Pedagogical Justification:** in the inertial frame the particle has cylindrical coordinates $\rho = L\sin\theta$, $z = -L\cos\theta$ and $\varphi = \omega t$, with $\dot{\varphi} = \omega$ constant, so $\ddot{\varphi} = 0$. No moving-frame theorem is needed: the acceleration comes from the cylindrical formulas (Notes Sec. 2.6.2).

$$\mathbf{r} = \rho\,\mathbf{e}_\rho + z\,\mathbf{k}, \qquad \mathbf{a} = \left(\ddot{\rho} - \rho\omega^2\right)\mathbf{e}_\rho + 2\dot{\rho}\omega\,\mathbf{e}_\varphi + \ddot{z}\,\mathbf{k}$$

- Differentiating with the chain rule $\frac{d}{dt}f(\theta) = f'(\theta)\dot{\theta}$:

$$\dot{\rho} = L\cos\theta\,\dot{\theta}, \qquad \ddot{\rho} = L\cos\theta\,\ddot{\theta} - L\sin\theta\,\dot{\theta}^2, \qquad \dot{z} = L\sin\theta\,\dot{\theta}, \qquad \ddot{z} = L\sin\theta\,\ddot{\theta} + L\cos\theta\,\dot{\theta}^2$$

3.2 Forces

- **Applied force:** the weight $-mg\,\mathbf{k}$, conservative.
- **Constraint force:** the reaction of the smooth track (or the string tension), normal to the circle. In part (a), $\mathbf{N} = -N\,\mathbf{e}_r$ with $N > 0$ pushing inward. In part (b) the reaction also has a component perpendicular to the plane of the circle: $\mathbf{N} = N_r\,\mathbf{e}_r + N_\varphi\,\mathbf{e}_\varphi$, with $\mathbf{N}\cdot\mathbf{e}_\theta = 0$.
- **Inertia forces:** none ($\mathcal{F}_0$ is inertial).
- In the basis $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ the weight is $-mg\,\mathbf{k} = mg\cos\theta\,\mathbf{e}_r - mg\sin\theta\,\mathbf{e}_\theta$.

3.3 Equations of Motion

Part (a)

> **Pedagogical Justification:** one unknown coordinate θ and one unknown reaction N, and two scalar equations in the plane. The tangential projection has no reaction and is the equation of motion; the normal projection gives N.

$$\mathbf{e}_\theta:\ mL\ddot{\theta} = -mg\sin\theta, \qquad -\mathbf{e}_r:\ N - mg\cos\theta = mL\dot{\theta}^2 = m\frac{v^2}{L}$$

- **Energy.** $\mathbf{N}\cdot\mathbf{v} = 0$ (the reaction is normal to the velocity), so the reaction does no work and the energy is a first integral. Multiplying the tangential equation by $L\dot{\theta}$ gives the same result. With the lowest point as the reference, $V(\theta) = mgL(1-\cos\theta)$:

$$E = \tfrac{1}{2}mv^2 + mgL(1-\cos\theta) = \tfrac{1}{2}mv_0^2 \implies v^2(\theta) = v_0^2 - 2gL(1-\cos\theta)$$

Part (b)

> **Pedagogical Justification:** projecting Newton's second law on the tangent $\mathbf{e}_\theta = \cos\theta\,\mathbf{e}_\rho + \sin\theta\,\mathbf{k}$ of the circle removes the reaction ($\mathbf{N}\cdot\mathbf{e}_\theta = 0$): $m\,\mathbf{a}\cdot\mathbf{e}_\theta = -mg\sin\theta$. Computing $\mathbf{a}\cdot\mathbf{e}_\theta = \left(\ddot{\rho} - \rho\omega^2\right)\cos\theta + \ddot{z}\sin\theta$:

$$\ddot{\rho}\cos\theta + \ddot{z}\sin\theta = L\ddot{\theta}\left(\cos^2\theta + \sin^2\theta\right) + L\dot{\theta}^2\left(-\sin\theta\cos\theta + \sin\theta\cos\theta\right) = L\ddot{\theta}, \qquad -\rho\omega^2\cos\theta = -L\omega^2\sin\theta\cos\theta$$

$$mL\ddot{\theta} - mL\omega^2\sin\theta\cos\theta = -mg\sin\theta \implies \boxed{\ddot{\theta} + \left(\frac{g}{L} - \omega^2\cos\theta\right)\sin\theta = 0}$$

- **Reaction components.** Perpendicular to the plane ($\mathbf{e}_\varphi$): $N_\varphi = m\cdot 2\dot{\rho}\omega = 2mL\omega\dot{\theta}\cos\theta$. In the plane, projecting on $\mathbf{e}_r = \sin\theta\,\mathbf{e}_\rho - \cos\theta\,\mathbf{k}$, with $\mathbf{a}\cdot\mathbf{e}_r = \left(\ddot{\rho} - \rho\omega^2\right)\sin\theta - \ddot{z}\cos\theta$. With $\rho = L\sin\theta$ and $z = -L\cos\theta$, $\ddot{\rho} = L\left(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta\right)$ and $\ddot{z} = L\left(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta\right)$, so $\ddot{\rho}\sin\theta - \ddot{z}\cos\theta = -L\dot{\theta}^2$ and $\mathbf{a}\cdot\mathbf{e}_r = -L\left(\dot{\theta}^2 + \omega^2\sin^2\theta\right)$:

$$N_r + mg\cos\theta = -mL\left(\dot\theta^2 + \omega^2\sin^2\theta\right) \implies -N_r = mL\left(\dot\theta^2 + \omega^2\sin^2\theta\right) + mg\cos\theta$$

For $\omega = 0$ this is the inward reaction of part (a).

3.4 Integration and Answers

Part (a): normal reaction and detachment

> **Pedagogical Justification:** the reaction follows from the normal equation with the energy relation, and the particle leaves the circle when N vanishes:

$$N(\theta) = mg\cos\theta + \frac{m}{L}\left[v_0^2 - 2gL(1-\cos\theta)\right] = \frac{mv_0^2}{L} - mg(2 - 3\cos\theta)$$

$$N(\theta_d) = 0 \implies \boxed{\cos\theta_d = \frac{2gL - v_0^2}{3gL}}$$

- **Regime 1, $v_0 \le \sqrt{2gL}$.** The speed vanishes when $v_0^2 = 2gL(1-\cos\theta_{\max})$, i.e. $\cos\theta_{\max} = 1 - \dfrac{v_0^2}{2gL} \ge 0$, so $\theta_{\max} \le \pi/2$. There $N(\theta_{\max}) = mg\cos\theta_{\max} \ge 0$: the particle stops and returns before N can vanish. **The detachment point is never reached**; the particle oscillates in the lower semicircle (at $v_0^2 = 2gL$ it reaches the horizontal at rest with $N = 0$ and turns back).
- **Regime 2, $\sqrt{2gL} < v_0 < \sqrt{5gL}$.** Since $v_0^2 > 2gL$ the particle crosses $\theta = \pi/2$ with positive speed, and $2gL < v_0^2 < 5gL$ gives $-1 < \cos\theta_d < 0$, i.e. $\theta_d \in (\pi/2, \pi)$. The speed there is $v^2(\theta_d) = v_0^2 - 2gL\,(1-\cos\theta_d) = \dfrac{v_0^2 - 2gL}{3} > 0$. **The detachment point is reached**: the particle leaves the circle at $\theta_d = \arccos\dfrac{2gL - v_0^2}{3gL}$ and continues in free (parabolic) flight under gravity.
- **Regime 3, $v_0 \ge \sqrt{5gL}$.** At the top, $N(\pi) = \dfrac{mv_0^2}{L} - 5mg \ge 0$, and $N(\theta)$ decreases monotonically from 0 to π, so the reaction stays non-negative along the whole loop. **The particle never leaves** the circle and makes complete turns.

Part (b): equilibrium positions

Pedagogical Justification: an equilibrium relative to the turning circle is a static solution of the equation of motion, $\ddot{\theta} = \dot{\theta} = 0$ (in the inertial frame the particle then describes a horizontal circle):

$$\left(\frac{g}{L} - \omega^2 \cos\theta\right) \sin\theta = 0$$

- **Vertical alignments** ($\sin\theta = 0$): $\theta_1 = 0$ (bottom) and $\theta_2 = \pi$ (top). The constraint of the statement is unilateral (the particle moves on the inside of the circumference). At rest, $-N_r = mL\left(\dot{\theta}^2 + \omega^2 \sin^2\theta\right) + mg\cos\theta$ gives $-N_r = mg > 0$ at the bottom (admissible) but $-N_r = -mg < 0$ at the top, an outward push that the inside of the hoop cannot exert. So $\theta_2 = \pi$ is an equilibrium only for a bilateral constraint (a bead on a wire) and is **excluded** here.
- **Off-axis equilibria** ($\cos\theta = g/(\omega^2 L)$): they exist if and only if $\left|g/(\omega^2 L)\right| \leq 1$, i.e. $\omega^2 \geq g/L$, and they are admissible because $-N_r = mL\omega^2 \sin^2\theta_0 + mg\cos\theta_0 > 0$ ($\cos\theta_0 > 0$). For $\omega > \sqrt{g/L}$ there are two symmetric positions $\theta_{3,4} = \pm \arccos \dfrac{g}{\omega^2 L}$.
- **Stability** (linearisation of $\ddot{\theta} = F(\theta)$ with $F = -\left(g/L - \omega^2\cos\theta\right)\sin\theta = -\frac{g}{L}\sin\theta + \frac{\omega^2}{2}\sin 2\theta$, so $F'(\theta) = -\frac{g}{L}\cos\theta + \omega^2\cos 2\theta$; at θ_0 with $\cos\theta_0 = g/(\omega^2 L)$, $F'(\theta_0) = -\omega^2\cos^2\theta_0 + \omega^2\left(2\cos^2\theta_0 - 1\right) = -\omega^2\sin^2\theta_0$): $F'(0) = -(g/L - \omega^2)$, so the bottom is stable for $\omega^2 < g/L$ and unstable for $\omega^2 > g/L$; near $\theta = \pi$, $\ddot{\theta} \approx (g/L + \omega^2)(\theta - \pi)$, always unstable; at $\theta_{3,4}$, $F'(\theta_0) = -\omega^2\sin^2\theta_0 < 0$, stable.

PHASE 4

Interpretation & Checks

- **Supercritical pitchfork:** for $\omega < \omega_c = \sqrt{g/L}$ the bottom is the only stable equilibrium. At ω_c it becomes unstable and two stable off-axis equilibria appear, at the angle where the horizontal circle of radius $L\sin\theta$ has the centripetal acceleration required by the balance with gravity. As $\omega \to \infty$, $\cos\theta_0 \to 0$ and $\theta_0 \to \pi/2$.
- **Limits of the regimes:** at $v_0^2 = 2gL$, $\cos\theta_d = 0$ ($\theta_d = \pi/2$), the boundary between oscillation and detachment; at $v_0^2 = 5gL$, $\cos\theta_d = -1$ ($\theta_d = \pi$) and $N(\pi) = 0$, the boundary between detachment and full loops. For $\omega \to 0$ the equation of (b) reduces to the simple pendulum $\ddot{\theta} + (g/L)\sin\theta = 0$.
- **Dimensional check:** $[N] = [mv_0^2/L] = \mathrm{N}$; $\cos\theta_d$ is dimensionless; $\omega_c = \sqrt{g/L}$ is in s^{-1}.

- **Check against the official key:** `Problems.pdf` has no entry for Problem 23 in its solution keys, so the results are verified by independent derivation. Numerical integration ($g = L = 1$, detachment found where $N = 0$) gives $\theta_d = 120.000^\circ$ for $v_0^2 = 3.5$ (with $v^2 = 0.5 = (v_0^2 - 2gL)/3$), 99.594° for $v_0^2 = 2.5$ and 165.165° for $v_0^2 = 4.9$, all equal to $\arccos\left[(2 - v_0^2)/3\right]$. Sympy confirms the equation of motion of part (b) and $N_\varphi = 2mL\omega\dot{\theta}\cos\theta$ from the inertial acceleration of the point on the turning circle.

Source: Problems.pdf